I-01-CX-frequency-scale-time-kill-test v1: Frequency-scale-time $\Rightarrow$ upper bound on W_+

ZFL-CX

September 3, 2026

1 Mission

$\boxed{T^* < \infty \Rightarrow C_I}$ $\boxed{C_E^{weak} \wedge C_{L^3} \wedge C_F^{conc} \wedge C_I \Rightarrow \perp}$ $\boxed{\text{Priority: } W_+ \leq f(C_I)}$ Goal $W(t) = \int \det(\nabla u)$, $W_+ = \max(W, 0)$. $C_F^{conc} : \|u(t_n)\|_{L^3(B_{r_n})} \geq \varepsilon_0$.

2 Definition C_I

$N_c(t) := \inf\{N : \|P_{>N}u(t)\|_{L^3} \leq \varepsilon_0/2\}$ $\boxed{C_I^{freq} : \exists \eta_0 > 0, \exists N_n \to \infty, t_n \uparrow T^*, \|P_{N_n}u(t_n)\|_{L^3} \geq \eta_0}$ $r_n = N_n^{-1}$.

3 Attempt $W_+ \leq f(C_I)$

$W = \sum N_1 N_2 N_3$ paraproduct. Attempt $|W| \lesssim \|u\|_{BMO} \|\nabla u\|_2^2$. C_I does not control BMO.

4 Kill

KILLED if C_I reformulation of C_{L^3}, only concentration $r_n \sim N_n^{-1}$ already in C_F^{conc}, reproduces ε-regularity, $W_+ \leq f$ with $f = +\infty$ possible, needs Type I, no W_+ link. All hold. No $W_+ \leq f(C_I)$ integrable.